

Project Initiation and Evaluation

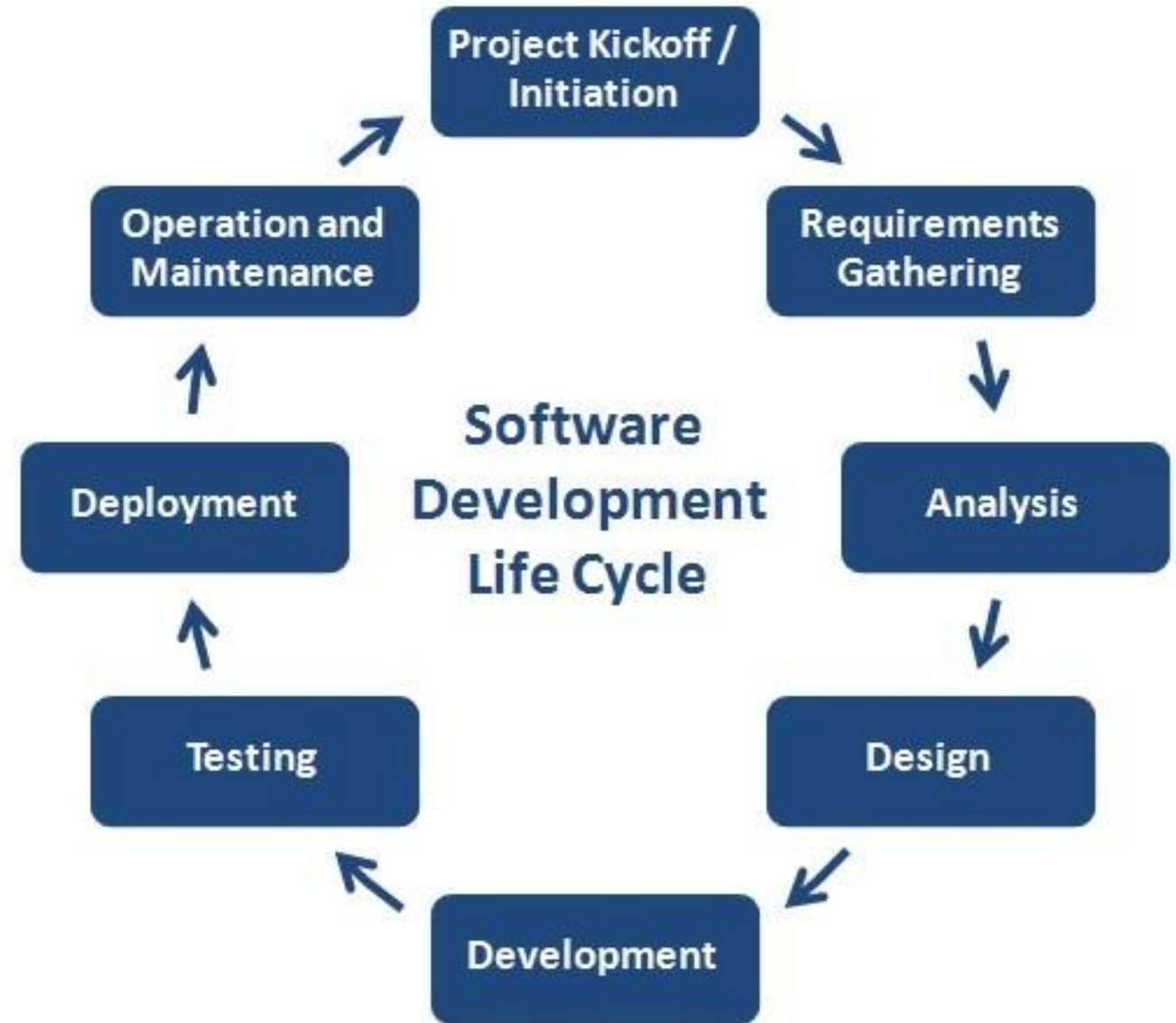
CSE 4032 - TOPIC 2

Mr. Madusanka Nipunajith

Agenda

- Project Initiation (Topic 1)
- Software Development Life Cycle
- Software Development Methodology
- Objectives in project initiation
- Activities in project initiation
- Project Evaluation (Topic 2)
- Assessments (3 Types) - Economic Assessment
- Cash Flow Forecasting
- Exercises

Phases Of Software Project Lifecycle



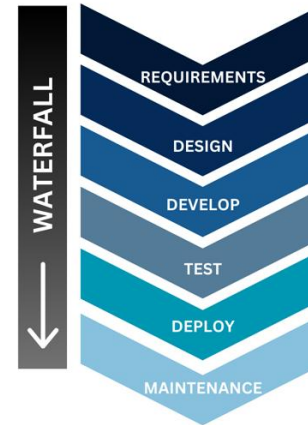
SDLC & Methodology

- The PDLC describes the sequential phases that every project passes through, regardless of approach.
- The PDLC represents what happens in a project from beginning to end - The natural progression of activities.
- Methodologies define how to approach the project work.
 - Waterfall
 - Agile
 - Hybrid
- The methodology is the framework and techniques for navigating through the life cycle phases.

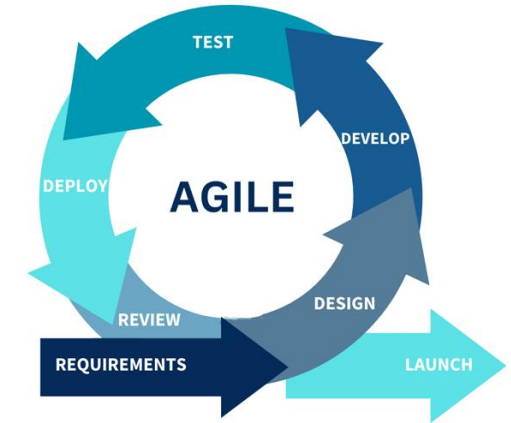


Software Development Methodologies

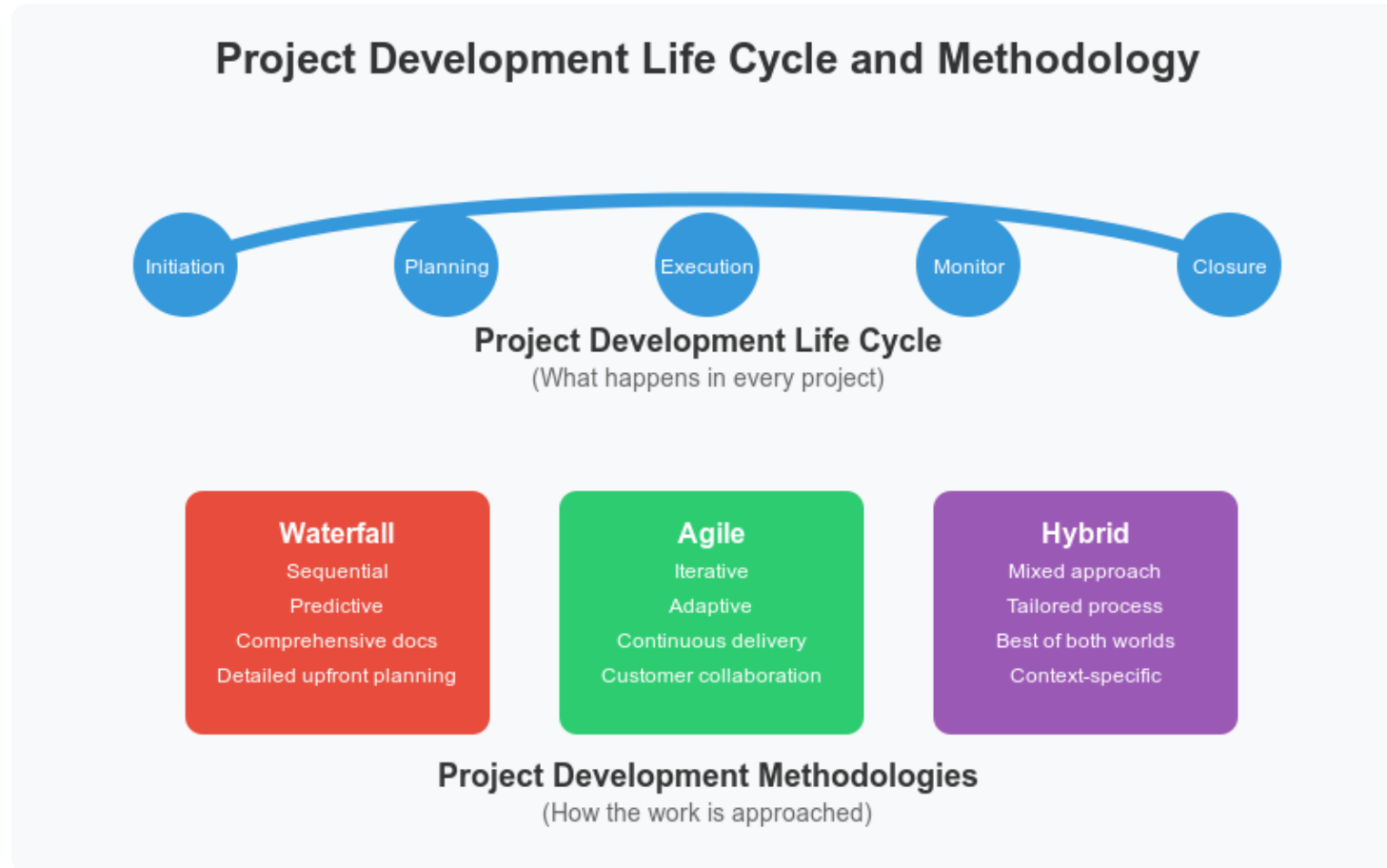
- Waterfall
 - Sequential, phase-based approach where each phase must be completed before the next begins.
- Agile
 - Iterative, incremental approach focusing on flexibility and customer collaboration.
 - Ex: Scrum, Kanban, XP (Extreme Programming)
- Hybrid
 - Combining elements of both waterfall and agile
 - Ex: SAgile (Scaled Agile Framework)



VS



SDLC & Methodology



Objectives in Project Initiation

- Align the stakeholders' expectations with the project purpose.
- Set the vision of the project – what needs to be accomplished.
- Provide visibility on scope and objectives.
- Reduce the overhead of involvement.
- Minimize risk and uncertainty.



Risk and Uncertainty

- Risk (Known Events)
 - Risk is something we know might happen, and we can plan for it.
 - Potential event
 - Example: Team member resigns during development
- Uncertainty (Unknown Events)
 - Uncertainty is when we don't even know what might happen or how to prepare.
 - completely unpredictable or immeasurable.
 - Example: A sudden new law that affects software usage (GDPR)



Handling Risk and Uncertainty

- Risk
 - Risk Identification (2 main approaches)
 - Risk Assessment (Risk Exposure)
 - Risk Mitigation Plan
- Uncertainty
 - Build flexibility into your plan
 - Agile methods
 - Continuous feedback loops
 - Keep a buffer budget/time for surprises



Activities in Project Initiation

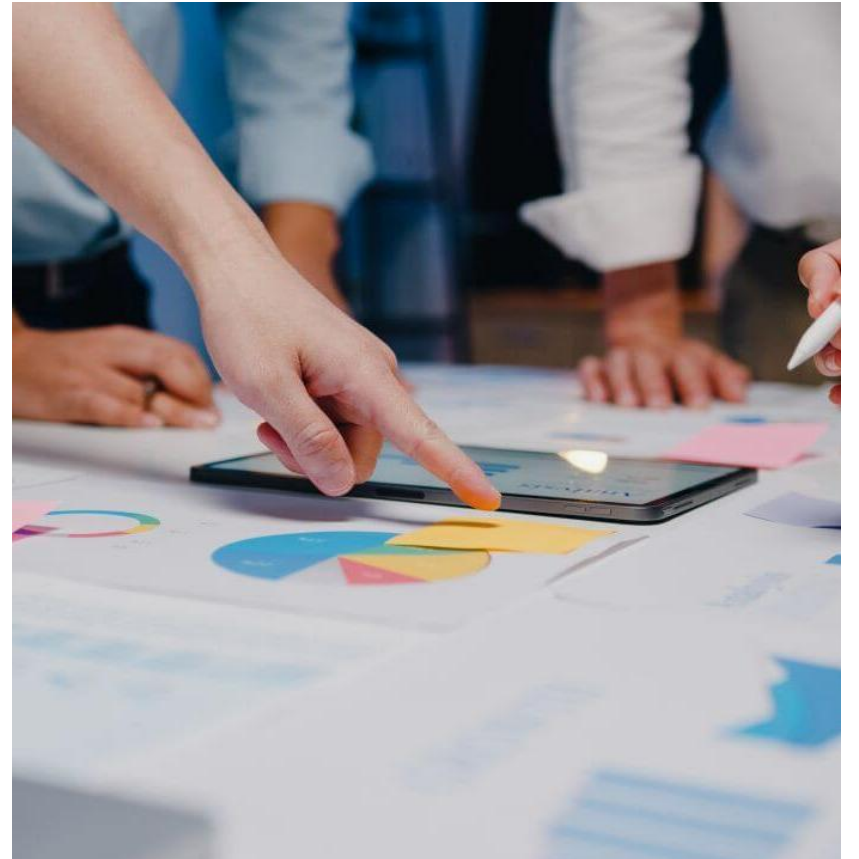
Project Initiation

Follow these six steps to start your project off right



Business Case Document

- Introduction and Background to the Proposal.
- The proposed project.
- The market: estimated demand, competitors.
- The benefits.
- The outline implementation plan.
- Costs.
- Risks.
- Management plan.



Business Case Document

Project Business Case Example			
Project Name	Sales Team IVR Telephone System		
Project Sponsor	Head of Sales	Project Manager	Name of project manager
Date of Project Approval	3rd March	Last Revision Date	3rd March
Contribution to Business Strategy	Our strategy is to project best in industry customer service, and the current situation does not reflect this. The new IVR system will ensure all calls are answered in a timely manner. It will also ensure that calls are delt with efficiently. These two facts align this project to the company strategy.		
Options Considered	Options considered included: 1. Adding additional staff to sales team 2. Having a dedicated team for our best customers 3. An IVR system (selected)		
Benefits	1. Increased sales - currently estimated we lose 4% of all sales calls due to current issues. 2. Happier customers - we estimate new customer satisfaction will increase by 10%. 3. Improved LTV - lifetime value of customers will increase by 5% due to the two points above		
Timescales	Initial analysis shows that the system will take approximately 3-4 months to implement.		
Costs	IVR software = \$35,000 Project Management = \$30,000 Software team of 3 for 3 months = \$90,000 Total estimated cost = \$155,000		
Expected Return on Investment	Year 1 = \$0 Year 2 = \$120,000 Year 3 = \$180,000 as LTV begins to be felt.		
Risks	Right now the project looks pretty straightforward but there are still some unknowns surrounding implementation. There is also the risk that the project doesn't meet the sales team or customers needs. For this reason it is recommended to involve the sales team closely.		

Feasibility Study

- A comprehensive analysis assessing the practicality and viability of a proposed project or business idea.
- Purpose: To decide whether the solution will deliver the expected outcome.
- Outcome: A confirmed solution for implementation.



Feasibility Study

FEASIBILITY STUDY

5 Types of Feasibility Studies

Legal Feasibility

Performed to understand if the proposed plan conforms to the legal and ethical requirements.

Economic Feasibility

Involves a cost benefits analysis to identify how well, or how poorly, a project will be concluded.

Technical Feasibility

Process of validating the technical resources and capabilities to convert the ideas into working systems.

Operational Feasibility

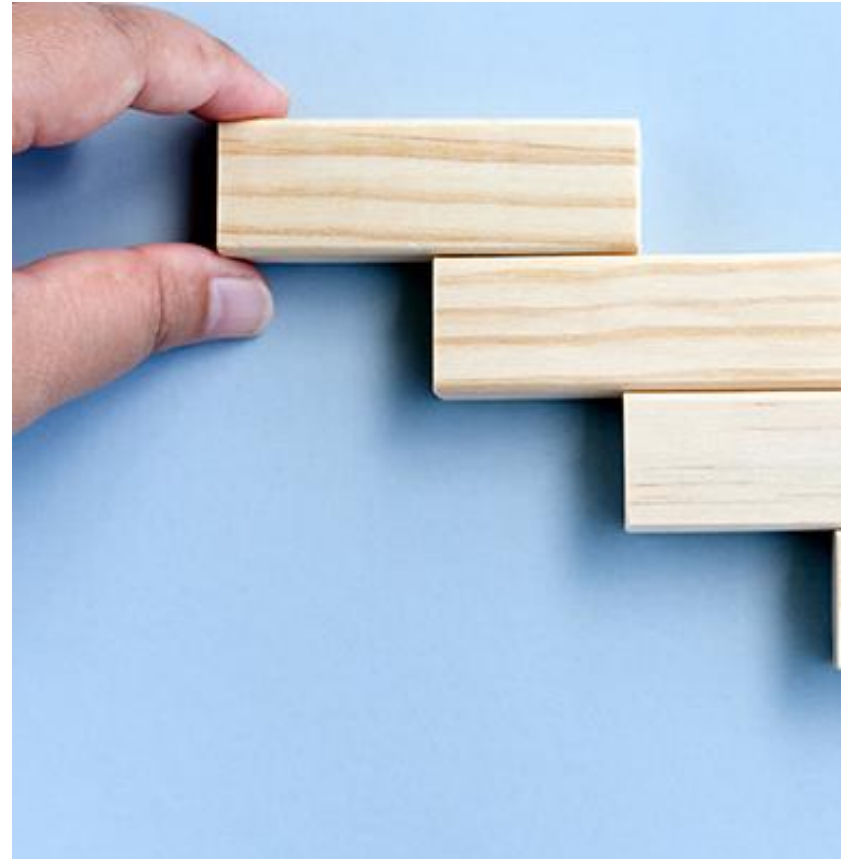
Performed to understand well a proposed system solves the problems.

Scheduling Feasibility

Measure of how reasonable the project duration is.

Steps to conduct the feasibility study

1. Describe the business problem or opportunity.
2. Document the business requirement for a solution.
3. Identify all of the alternative solutions available.
4. Review each solution to determine its feasibility.
5. List any risks and issues with each solution.
6. Choose a preferred solution.
7. Document the result.



Steps to conduct the feasibility study

1. Describe the business problem or opportunity.
 - Website crashes when more than 100 users access it simultaneously.
2. Document the business requirement for a solution.
 - System must handle 500 concurrent users without performance issues.
3. Identify all of the alternative solutions available.
 - Upgrade current server, move to cloud hosting, or implement load balancing.
4. Review each solution to determine its feasibility.
 - Cloud hosting can scale automatically to meet traffic demands with minimal configuration.
5. List any risks and issues with each solution.
 - Cloud migration might cause 4-hour downtime; staff needs training on new platform.
6. Choose a preferred solution.
 - Select AWS cloud hosting for its scalability and established migration tools.
7. Document the result.
 - AWS cloud implementation will take 3 weeks and cost \$5,000 for the migration.



Project Charter



Defines the purpose
(goals and objectives)
of the project.



Decides the project
duration.



Identifies the project
scope and
deliverables.



Identify the financial
and other resource
requirements.



Identifies the
stakeholders and their
roles and
responsibilities.

Project Charter

Project Charter Template							
PROJECT TITLE		Employee and Rewards recognition Application		DESCRIPTION			
				Automate and simplify rewards and recognition program with easy-to-use and customizable cloud-based solution.			
BUSINESS NEED			CORE TEAM MEMBERS		KEY STAKEHOLDERS		
Build a cost-effective rewards and recognition program that is scalable, customizable, and meets your company's goals.			Name	Role	Name	Role	
Improve organization ability to attract and recruit high quality talent and compete in the Market			Mark Tim	Project Manager	Microsoft	Sponsor	
Component of healthy Workplace			Vijay Kshyap	Team Leader	Investors	Shareholders	
Element of Performance Management			Neelam	Business Analyst	Users	Users	
Overall create healthy workplace and retention of quality talent			Nina	Graphic Designer			
OBJECTIVES							
To create a friendly environment and healthy place where resources would enjoy working in the organization and maintain long term professional relationship							
To recognize resource for their commendable and high quality service and commitment							
Automate and simplify your rewards and recognition program with our easy-to-use and customizable cloud-based solution.							
GOALS			SUMMARY PROJECT STATUS				
The goal of the project is to retain and hire high quality resources. Set standards for employees so they have concrete goals to reach and encourage resources to set goals that are Specific, Measurable, Attainable, Relevant and Timely – that provides employees a clear path to follow and gives their supervisors established metrics to measure progress.			Project Start Date:		2021-04-15		
			Project Completion Date:		2021-09-15		
			Overall process Impacts:		New Customizations or Changes		
			Potential Financial Budget:		\$100,000,000		
SCOPE		OPPORTUNITY	MILESTONES		STATUS	DUE	
The scope is to create a new Employee and Rewards recognition full fledged Application where user and supervisor measure their respective progress		Currently, Microsoft employee recognition portal doesn't help to measure progress neither encourages high quality resources. Microsoft foresees an opportunity to improve their HR processes so that they can attract the best talent available in the Market	Requirement Gathering		In Progress	2021-04-17	NA
			Current Business Process Analysis		In Progress	2021-04-26	NA
			Project Presentation		Not Started	2021-06-25	NA
			Future Business innovations		Not Started	2021-07-25	NA
			Performance Analysis		Not Started	2021-08-04	NA
			Final Application Delivery		Not Started	2021-09-09	NA
CONSTRAINTS			ASSUMPTIONS				
1) Time Constraint 2) Budget Constraint 3) Resource Constraint			1) We assume that the above estimation is based on the consideration that would be no major changes 2) We assume resources be available as required else there would be delay in overall deliveries 3) We assume stakeholders active participation else there would be a delay in deliveries				
DELIVERABLES							
1) Wireframes and prototype explaining the flow of the applications 2) Deliverables will be pushed every 3-4 weeks 3) Licences and plugin required will be provided by Sponsor 4) Hosting services will be purchased and provided by the client 5) IT Department will test new security and privacy settings.							
SPONSOR NAME			SIGN & DATE				
MATTHEW WATSON							

Business Case vs. Project Charter

- Simple Analogy
 - Business Case: Should we do this project and why? (If yes, Project Charter)
 - Project Charter: We are doing this project. Here's how and who's responsible
- Business Case (Hospital System)
 - The hospital needs a new patient record system to reduce paperwork, improve accuracy, and save 20% in admin costs. The investment is LKR 4M, with an expected ROI of 35% in 2 years.
- Project Charter (Hospital System)
 - This project will develop and deploy a new patient record system. The project manager is Madusanka, and the project will include features A, B, and C. It must be delivered in 6 months under an LKR 4M budget.

Business Case vs. Project Charter

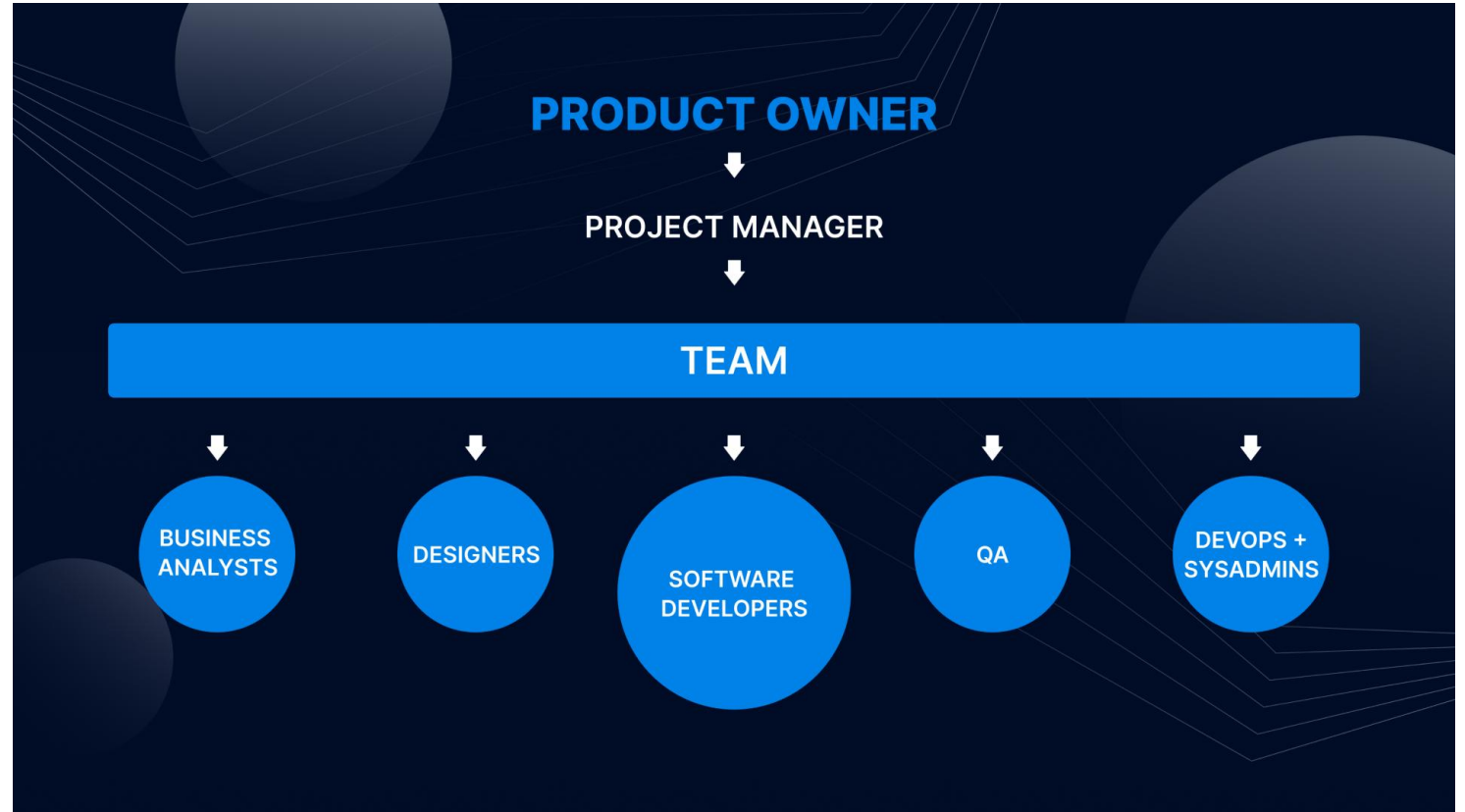
- Exercise (10 mins)
 - What is the difference between a project charter and a business case in different aspects?
 - Purpose
 - Focus
 - Content
 - Created by
 - Audience
 - Timing
 - Outcome



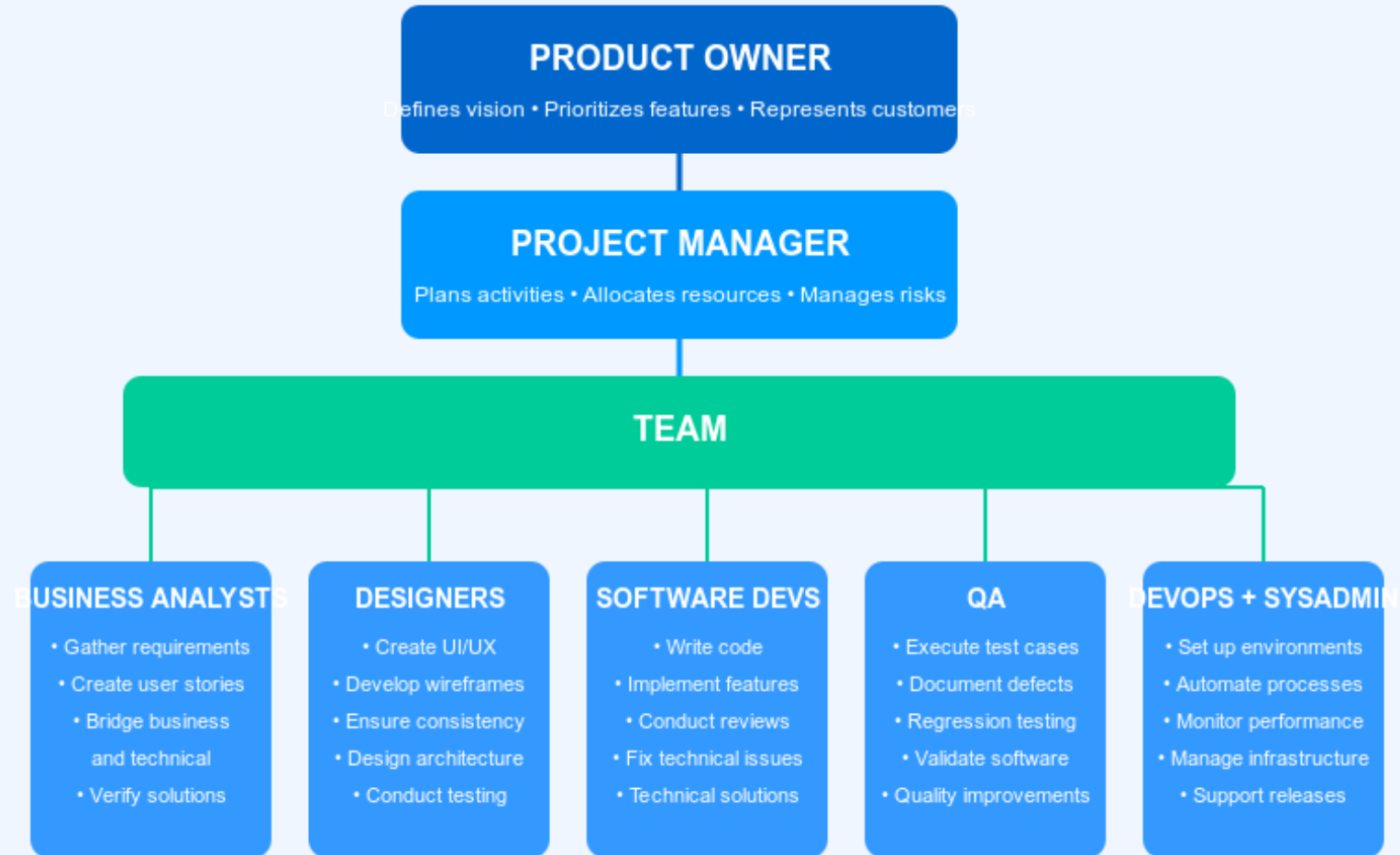
Business Case vs. Project Charter

Aspect	Business Case	Project Charter
Purpose	Justify why the project should be undertaken	Formally authorizes the project to begin
Focus	Value, Benefits, ROI and etc	Scope, Authority, Responsibilities and etc
Content	Cost benefit analysis, Risks and etc	Objectives, Stakeholders, Constraints and etc
Created by	BA (Business Analyst)	Product/Project Manager
Audience	Executives, Decision makers	Project team and stakeholders
Timing	Before the project is approved	After the project is approved
Outcome	Leads to go/no-go decision	Grants formal authorization to start the project

Project Team



Software Development Team Responsibilities



Effective software delivery requires clear responsibilities and collaboration

Project Management Office

- Organizing meetings and events
- Resource acquisition and allocation
- Support monitoring and controlling
- Communication handling
- Administrative record handling
- Reporting lessons learnt
- Prepare project reports

